



Your Submission MATCOM-D-26-01427

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Date Sun 6/14/2026 6:41 AM

To Heather Jiles <hjiles@gmu.edu>

 2 attachments (179 KB)

Review Report on MATCOM-D-26-01427.pdf; MATCOM-D-26-01427-Report.pdf;

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CC: "Robert Weigel" rweigel@gmu.edu

Ms. Ref. No.: MATCOM-D-26-01427

Title: High-Accuracy Numerical Solutions of Particle Motion in Static Magnetic Fields
Mathematics and Computers in Simulation

Dear Dr.Ms. Heather Jiles,

Comments on your work have now been received. On the basis of these reviews your Editor is advising against publication of your manuscript. Therefore, I must reject it.

For your guidance, all comments are appended below or can be downloaded.

Thank you for giving us the opportunity to consider your work.

Yours sincerely,

Rosa Maria Spitaleri, Research Director
Editor in Chief
Mathematics and Computers in Simulation

Reviewers' comments:

Reviewer #1: Please see the attached file.

Reviewer #3: This manuscript presents a study of the Parker-Sochacki (PS) method against several Runge-Kutta (RK) type integrators (RK4, adaptive RK45, and symplectic Gauss-Legendre RKG) for solving the Lorentz equations of charged particles in static magnetic fields. Three magnetic field configurations are considered: uniform, hyperbolic tangent, and dipole. The authors demonstrate that the PS method achieves significantly lower relative kinetic energy errors (4-13 orders of magnitude improvement) and better long-term stability than the RK methods. The paper also includes extensive numerical experiments, error-matched comparisons, and validation against

analytical bounce/drift periods. The work is technically sound, well-structured, and contains a substantial amount of computational evidence. However, the following issues should be addressed before publication.

1. Some key parameters are not fully specified. (a) How is the adaptive truncation criterion implemented in practice? What is the default tolerance? (b) For the "tethering" strategy, the authors reinitialize auxiliary variables analytically. Does this introduce any inconsistency with the truncated series representation?
2. The RKG method fails for all electron configurations in the dipole field. The authors cite Yugo & Iyemori (2007), who noted similar issues. A brief explanation of this failure would help readers understand the robustness of the PS method.
3. Several sentences are long and complex, especially in Sections 2–3. The manuscript would benefit from breaking up these sentences to improve clarity while preserving all technical content.
4. The paper strongly advocates the PS method but hardly mentions scenarios where PS might be less suitable than RK methods. Adding a brief "Limitations" paragraph would provide a more balanced perspective.

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